OPEN ACCESS
International Journal of Neural Systems, Vol. 32, No. 7 (2022) 2250031 (13 pages)

DOI: 10.1142/S0129065722500319

Fractal Dimension Feature as a Signature of Severity in Disorders of Consciousness: An EEG Study

Camillo Porcaro*
Department of Neuroscience and Padova Neuroscience Center (PNC)
University of Padova, Padova 35128, Italy
Institute of Cognitive Sciences
and Technologies (ISTC) — National Research Council (CNR)
00185 Rome, Italy
Centre for Human Brain Health, School of Psychology
University of Birmingham, Birmingham, B15 2TT, UK
**camillo.porcaro@unipd.it*

Marco Marino
Movement Control and Neuroplasticity Research Group
KU Leuven, Leuven, Belgium
IRCCS San Camillo, Hospital, Venice, Italy

Simone Carozzo, Miriam Russo, Maria Ursino, Valentina Ruggiero
Carmela Ragno, Stefania Proto and Paolo Tonin
S.Anna Institute — Research in Advanced
Neurorehabilitation, 88900 Crotone, Italy

Accepted 22 April 2022
Published Online 7 July 2022

An accurate diagnosis of the disorder of consciousness (DOC) is essential for generating tailored treatment programs. Accurately diagnosing patients with a vegetative state (VS) and patients in a minimally conscious state (MCS), however, might be very complicated, reaching a misdiagnosis of approximately 40% if clinical scales are not carefully administered and continuously repeated. To improve diagnostic accuracy for those patients, tools such as electroencephalography (EEG) might be used in the clinical setting. Many linear indices have been developed to improve the diagnosis in DOC patients, such as spectral power in different EEG frequency bands, spectral power ratios between these bands, and the difference between eyes-closed and eyes-open conditions (i.e. alpha-blocking). On the other hand, much less has been explored using nonlinear approaches. Therefore, in this work, we aim to discriminate between MCS and VS groups using a nonlinear method called Higuchi's Fractal Dimension (HFD) and show that HFD is more sensitive than linear methods based on spectral power methods. For the sake of completeness, HFD has also been tested against another nonlinear approach widely used in EEG research, the Entropy (E). To our knowledge, this is the first time that HFD has been used in EEG data at rest to discriminate between MCS and VS patients. A comparison of Bayes factors found that differences between MCS and VS were 11 times more likely to be detected using HFD than the best performing linear method tested and almost 32 times with respect to the E. Machine learning has also been tested for HFD, reaching an accuracy of 88.6% in discriminating among VS, MCS and healthy controls. Furthermore, correlation analysis showed that HFD was more robust to outliers than spectral power methods, showing a clear

*Corresponding author.

positive correlation between the HFD and Coma Recovery Scale-Revised (CRS-R) values. In conclusion, our work suggests that HFD could be used as a sensitive marker to discriminate between MCS and VS patients and help decrease misdiagnosis in clinical practice when combined with commonly used clinical scales.

Keywords: Disorder of consciousness (DOC); vegetative state (VS); minimally conscious state (MCS); Higuchi's fractal dimension (HFD); entropy (E); electroencephalography (EEG); coma recovery scale-revised (CRS-R).

1. Introduction

The definition of consciousness still represents one of the most challenging and open questions in neuroscience.[1] Many theories suggest that this does not depend upon the activity of a specific brain region but rather arises from complex networking among many different regions.[2] Consciousness is clinically classified in wakefulness (i.e. the presence of spontaneous periods of opening the eyes) and awareness (i.e. the ability of a subject to respond to the internal/external stimuli). Based on these two classifications, patients with disorders of consciousness (DOC)[3] might be subdivided into different states of consciousness such as coma, Vegetative State (VS) or Unresponsive Wakefulness Syndrome (UWS), and Minimal Conscious State (MCS). VS term refers to patients who appear to be wakeful with cycles of closing and opening of their eyes (full arousal) but with no sign of purposeful response (no awareness),[4] while MCS refers to patients with inconsistent weak, voluntary responses.[5]

Diagnosis of DOC is limited to the clinical evaluation of motor responses mainly impaired or undetectable for VS and MCS patients.[6] Among many different scales available in the literature, the Coma Recovery Scale-Revised (CRS-R)[7] is the most used. Correct use of this scale significantly ameliorated the reliability of the diagnosis in the clinical routine. Nevertheless, the complexity of the behavioral assessment in this kind of patient still is reflected by a high percentage of misdiagnosis between MCS and VS patients.[8,9] The high rate of misdiagnosis represents a critical challenge for neurologists to improve prognosis and selection of treatments and facilitate end-of-life decisions.[10,11]

Therefore, the use of an electrophysiological tool such as electroencephalography (EEG), by detecting the brain's electrical activity, might provide neurological biomarkers that may help address this issue and decrease the misdiagnosis rate.

EEG has some advantages to neuroimaging techniques such as functional Magnetic Resonance Imaging (fMRI) and Positron Emission Tomography (PET) techniques. First, EEG provides a direct measure of neuronal activity of the brain, while fMRI and PET only indirect measures. Moreover, EEG is costless and suitable for use at the patient bedside. Thanks to those advantages, the EEG community developed many different indexes to characterize the EEG data in healthy and pathological conditions when data are recorded at rest.[12–17] The most used is power spectrum density (PSD), estimated among different EEG frequency bands.[17–19] In particular, theta (θ), and alpha (α), bands are the most used since those are involved in human cognition.[20] As well, the ratio between alpha and theta band (α/θ), and the alpha band ratio between open and closed eyes, also known as alpha-blocking (α*Blocking*), are used in discriminating different neurodegenerative conditions[21] and consciousness.[22]

However, despite linear methods being predominantly used in characterizing brain oscillations in both healthy and pathological conditions, linear analysis may not be suitable to describe irregular and non-periodic patterns recorded by electrophysiological techniques.[23,24] Electrophysiological processes are generally nonstationary and nonlinear by nature.[25,26] Moreover, it has been recently shown that in many cases, the brain signals do not represent a sustained oscillation at a particular frequency but rather represent brief bouts of activity that repeat intermittently (*non-rhythmic*).[27,28] Acknowledging that physiological time series might contain *hidden information* captured by nonlinear methods such as *fractal analysis* might shed light on the physiological neural communication, computation, and cognition involved during healthy and pathological conditions so far overlooked by linear methods.[29–34] This might be the reason why time-series fractal approaches are more and more used in different research fields ranging from basic neuroscience,[35,36] neurophysiology,[37–41] translational neuroscience[16,42,43] to genetic variability in human phenotypes.[44–46]

To this end, we characterize the specific EEG signature at rest, using a time-series fractal approach called Higuchi's Fractal Dimension (HFD).[47] We then compare the results obtained with the abovementioned indexes estimated by classical linear methods, such as the well-known fast Fourier transformation (FFT) that remain a good choice if the analyzed signals are stationary and a different nonlinear measures such as HFD and Entropy[48] (E).

In this study, we aimed to apply linear (PSD in δ, τ, α, β, and γ bands plus α/θ and $\alpha Blocking$) and nonlinear (HFD, E) methods to the EEG recordings at rest with the aim to test whether nonlinear methods are more sensitive to discriminate between MCS and VS conditions, and consequently to provide an objective and effective electrophysiological marker for helping diagnosis in DOC patients. To this end, these electrophysiological linear and nonlinear markers were correlated with CRS-R clinical scale.

2. Materials and Methods

2.1. *Participants*

Ten healthy controls (HC) subjects (age 48.1 (mean) $\pm$ 4.84 (standard deviation), 6 female) and 17 DOC patients, 10 diagnosed as MCS (age 57.7 $\pm$ 12.9, 5 female) and seven as VS (age 57.1 $\pm$ 14.6, 3 female) were enrolled (Table 1).

Table 1. Demographics information. M: Male; F: Female; HEM: Hemorrhage; TBI: Traumatic Brain Injury; ISC: Ischemic; MCS: Minimally Conscious State; VS: Unresponsive Wakefulness Syndrome/Vegetative State; HC: Healthy Controls; CRS-R: Coma Recovery Scale-Revised; SD: Standard Deviation.

	HC	**MCS**	**VS**
#Subjects	10	10	7
Gender			
Male	4	5	4
Female	6	5	3
Age			
Mean (SD)	48.1(4.84)	57.7 (12.9)	57.1 (14.6)
CRS-R			
Mean (SD)		10.4 (1.43)	6.43 (0.787)
Range		[9, 13]	[6, 8]
Hospitalization (Days)			
Mean (SD)		94.5 (35.6)	162.0 (86.4)
Etiology		7: HEM 2: ISC 1: TBI	2: HEM 2: ISC 3: TBI

The consciousness diagnosis was based on clinical consensus and CRS-R assessment. The enrolled patients were hospitalized in a special rehabilitation unit for patients with severe DOC at the S. Anna Institute of Crotone (Italy). The inclusion criteria were: (1) age more than 16, (2) no administration of neuromuscular blockers or sedation within 24 h of enrolment, (3) eyes opening (indicating wakefulness and rest cycles), (4) diagnosis of VS or MCS, based on behavioral assessments by way of CRS-R,[49] (5) stable clinical condition. Exclusion criteria were: (i) persistence of craniectomy after brain injury; (ii) previous psychiatric or neurologic disorders; (iii) administration of pharmacological drugs interacting with the level of consciousness. This study was conducted after approval of the Ethical Committee of "Regione Calabria Comitato Etico Sezione Area Centro" n.ro 172 17–July–2020. The surrogate decision-makers of the patients enrolled in the study provided their written informed consent. The original forms were collected and stored. All the experimental procedures were conducted according to the policies and ethical principles of the Declaration of Helsinki.

2.2. *EEG recording*

EEG data were acquired at 1024 Hz by the Neurowave device (http://khymeia.com/prodotti/neurowave/). A 32 channel cap with sintered Ag–AgCl ring electrodes was used for the EEG recording, and two polygraphy channels for the Electrocardiogram (ECG) and the Electrooculogram (EOG). Recording impedances were kept below 5 kΩ. All signals were recorded with common reference; the signal ground was connected to an electrode on the left arm.

2.3. *EEG pre-processing*

Before the offline analysis, the data were downsampled to 256 Hz and bandpass filtered (1–48 Hz). An independent component analysis (ICA)-based procedure was implemented to identify and remove artefacts from data.[50,51]

2.4. *Characterization of electrophysiological neural activity at rest*

We investigated the electrophysiological properties of the EEG data from three different angles: (1) *Frequency domain* — PSD (i.e. the squared modulus of the continuous Fourier transform) is a particularly useful tool for studying brain oscillations on a time scale of minutes, which is typical of an individual's "stable state",[52] (2) *Time-domain* — signal power of neuronal assemblies, as a function of frequency, displays a "power law" function,[53] and the exponent of this function corresponds to its fractality. Thus, we used temporal Higuchi's fractal dimension (HFD)[47] as a signature of neural dynamics underlying brain functions, (3) *Signal distribution domain* — Entropy (E) introduced by Shannon in 1948[54] is a measure of uncertainty or unpredictability of a signal. It is defined as the expectance of the logarithm of the corresponding distribution probabilities. Entropy quantifies how different the distribution is from a uniform one. Therefore, we can use entropy to quantify to what degree a frequency component is fluctuating from a stationary process.[48]

Both HFD and E can be seen as complexity measures of the signal, even if they use a different property from the signal itself.

2.4.1. *Power spectral density (PSD)*

We calculated the PSD using the Welch procedure (256 ms duration, Hanning window, and 60% overlap). We then investigated the spectral properties in the classical frequency bands, such as δ (1–3 Hz), τ (4–7 Hz), α (8–13 Hz), β (14–30 Hz) and γ(31–45 Hz) bands.[55] In addition, we also calculated other two indexes derived from PSD estimates such as the ratio between α and θ bands (α/θ)[21] and the alpha band ratio between open and closed eyes (known as αBlocking).[22] Before the PSD estimation, the data were normalized to zero mean and unit variance. All the indexes refer to the total power value if not differently indicated.

2.4.2. *Higuchi's fractal dimension (HFD)*

HFD[47] is a nonlinear measure of waveform complexity applied in the time domain. Discretized functions or signals can be analyzed as a segment of data $X(1), X(2), \ldots, X(N)$, where N is the total number of samples. From the starting time sequence, a new self-similar time series X_m^k can be calculated as

$$X_m^k : x(m), \quad x(m+k), \quad x(m+2k), \ldots, x \times \left(m + \mathrm{int}\left(\frac{N-k}{k}\right)k\right) \tag{1}$$

for $m = 1, 2, \ldots, k$ where m is the initial time; k is the time interval, $k = 1, 2, \ldots, k_{\max}$; $k_{\max}$ is a free parameter, and $\mathrm{int}(r)$ is the integer part of the number r.

The length, $L_m(k)$, of each curve X_m^k is calculated as

$$L_m(k) = \frac{1}{k}\left[\sum_{i=1,\mathrm{int}(\frac{N-m}{k})} |X(m+ik) - X(m+(i-1)k| \frac{N-1}{\mathrm{int}(\frac{N-m}{k})}\right], \tag{2}$$

where N is the length of the original time series X and $(N-1)/\{\mathrm{int}[(N-m)/k]k\}$ is a normalization factor. $L_m(k)$ was averaged for all m forming the mean value of the curve length $L(k)$ for each $k = 1, \ldots, k_{\max}$ as

$$L(k) = \frac{\sum_{m=1}^{k} L_m(k)}{k}. \tag{3}$$

An array of mean values $L(k)$ was obtained and the HFD was estimated as

$$\mathrm{HFD} = \ln(L(k))/\ln(1/k) \quad \text{for } k = 1, 2, \ldots, k_{\max}. \tag{4}$$

In practice, the original curve or signal can be divided into smaller parts with or without overlap, called "windows". Then, the method for computing the HFD should be applied to each window, where N should be seen as the length of the window. In that case, HFD values are calculated for each window, with or without overlap. Individual HFD values can be averaged across all windows for the entire curve (or data time-series), and the mean HFD value can be used as a measure of curve complexity. Additional analysis demonstrated that HFD measurements were not dependent on the choice of window length and overlapping windows see Refs. 13, 14 and 32 for details. Here, for each EEG channel, we calculated HFD in non-overlapping time windows of 1 s (corresponding to 256-time points since our sample frequency rate was 256 Hz) as a good compromise between windows length of the data and computational time. The choice of the free parameter k has a crucial role in HFD estimation; for this reason, for each window, we estimated 127 values of FD for all the possible k values (i.e. $k = 2, \ldots, 128$).

The value 128 was equal to half of the samples within our 1 s window (i.e. 128-time points the maximum that can be chosen since the maximum k value is equal to half of the window length). For the subsequent HFD analysis, we set $k = 46$, which corresponds to the HFD median value for HC and PS groups[13,14,32] (see Fig. S.1). The value used for the analysis of variance (ANOVA) refers to the HFD estimated for each channel and averaged across all the channels afterwards.

2.4.3. *Entropy* (E)

E has been proposed as a measure of variability or uncertainty of a given random variable. E is defined as

$$E = -\sum_{i=1}^{n} P_i \log P_i. \tag{5}$$

Theoretically, E reaches its global maximum when $\{P_i\}$, the probability distribution function (pdf) of the variable under investigation, is uniformly distributed, for example, $P_i \neq P_j$, $i \neq j$. In the case of the measured signals, the pdf is unknown and should be estimated. An established method for estimating the pdf is the histogram method, in which the amplitude range of the signal is linearly divided into k bins so that the ratio k/N is constant (N is the number of signal samples). The ratio k/N characterizes the average filling of the histogram. The Freedman–Diaconis rule was applied to estimate the optimal number of bins.[16,56–58] The value used for the analysis of variance (ANOVA) refers to the entropy estimated for each channel and averaged across all the channels afterwards.

2.5. *Statistical analysis*

Shapiro–Wilk test for normality revealed that PSD, HFD and E values were different from Gaussian distribution ($p < 0.05$). We applied a logarithmic transformation and we obtained that $\log_{10}$(PSD), $\log_{10}$(HFD) and $\log_{10}(E)$ did not differ from a Gaussian distribution ($p > 0.200$). Repeated-measures analysis of variance (rm-ANOVA) was performed on PSD values to investigate the interaction effect GROUPs $\times$ BANDs (the three GROUPs as a between-subject factor: HC, VS and MCS; the five BANDs plus the α/θ ratio as a within-subjects factor (δ, θ, α, β, γ, and α/θ). The sphericity of the covariance matrix was verified with the Mauchly sphericity test. In the case of violation of the sphericity assumption, the Greenhouse–Geisser epsilon adjustment was used. One-way ANOVA was also applied to investigate the GROUPs effect (between-subject factor: HC, VS and MCS) on HFD, E and the α*Blocking* (as dependent variables). For all ANOVA models, the results were analyzed only if the Wilks' Lambda multivariate significance criterion was achieved. In the case of violation of the sphericity assumption, the Greenhouse–Geisser epsilon adjustment was used. *Post-hoc* analysis was performed using the Bonferroni correction method for multiple comparisons. Cohen's d value was used as a measure of effect size to measure the strength of the relationship between variables in a population, and it showed to be very useful when the numerosity of the data is low — as in this case, especially for the VS group. Finally, the three measures (PSD, HFD and E) were analyzed using a Bayesian approach to test which of the three methods were better able to discriminate between MCS and VS conditions. Analysis of Spearman's correlation coefficient was performed, respectively, between linear (PSD) and nonlinear (HFD and E) measures and CRS-R. All the analyses described above were conducted in JAMOVI software (v1.6.15 - jamovi.org/).

The single channel analysis was investigated by non-parametric statistical analysis, which does not assume Gaussian distribution of signals. Specifically, sample permutation t-tests (10,000 permutations; $p < 0.05$) between EEG channels and HC versus PS groups and between MCS versus VS groups. False discovery rate (FDR) was used to correct for multiple comparisons across channels.[59]

2.6. *Machine learning approach*

A Support Vector Machine (SVM) approach was tested using the 1 second window (for details see Sec. 2.4.2. Higuchi's fractal dimension (HFD) above) for a total of 761 values subdivided as follows: 245 values for HC, 234 values for MCS and 282 values for VS. The performance of SVM classifier was investigated as implemented in Matlab 2021a classification learner (*https://nl.mathworks.com/help/stats/classificationlearner-app.html*). The hyperparameters of the classifier (left as default) were optimized to minimize the five-fold cross-validation methods for protecting against overfitting by partitioning the data set into 5-folds and estimating accuracy on each fold.

3. Results

3.1. *Demographic results*

No significant age difference emerged among groups ($F(2, 24) = 2.24$, $p = 0.129$) by using a one-way ANOVA model after testing for Normality (Shapiro–Wilk test ($p = 0.242$)).

3.2. *Power Spectrum Density (PSD)*

An rm-ANOVA for the PSD log-transformed with a Greenhouse–Geisser correction (Mauchly's $W = 0.0183$, $p < 0.001$, $\varepsilon = 0.487$) revealed that the interaction effect GROUPs $\times$ BANDs was significant ($F(10, 120) = 15.7$, $p < 0.001$). The between-subjects factor showed a trend difference among the GROUPs ($F(2, 24) = 3.17$, $p = 0.060$). Post-hoc tests using the Bonferroni correction revealed that alpha and beta bands were significantly higher for the HC to the VS group ($p = 0.003$) (Fig. 1). Moreover, the α/θ ratio was also significantly higher for the HC to the MCS and the VS groups ($p < 0.001$ and $p < 0.001$, respectively) in both cases, but no significance between MCS and VS groups was reached (see Fig. 1) and Fig. S.2.

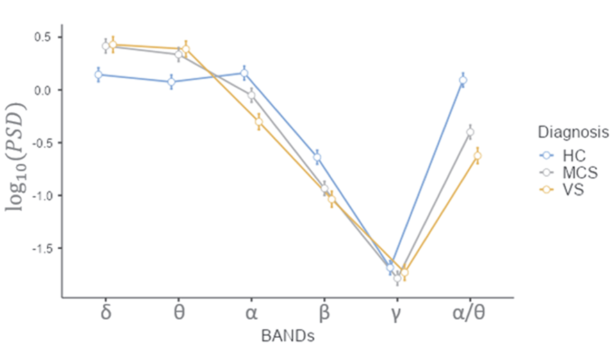

Fig. 1. PSD log-transformed in the Delta (δ), Theta (θ), Alpha (α), Beta (β), Gamma (γ), and Alpha and Theta ratio (α/θ) bands comparisons among diagnosis for HC, MCS, and VS groups.

Finally, a statistically significant difference between GROUPs as determined by the one-way ANOVA model was found for the α*Blocking* log-transformed values ($F(2, 24) = 21.1$, $p < 0.001$). A Bonferroni corrected *post-hoc* test revealed that α*Blocking* values were statistically significantly lower for the MCS (mean: $0.151 \pm$ standard error: 0.0385, $p < 0.001$, Cohen's $d = 2.467$) and VS (0.117 ± 0.0460, $p < 0.001$, Cohen's $d = 2.743$) compared to the HC group (0.451 ± 0.0385). No difference was observed for the MCS and VS groups ($p = 0.842$, Cohen's $d = 0.277$) see also Fig. 2 (upper panel) and Fig. S.2.

3.3. *Higuchi's Fractal Dimension (HFD)*

HFD log-transformed showed a statistically significant difference between *DIAGNOSIS* as determined by the one-way ANOVA model ($F(2, 24) = 28.7$, $p < 0.001$). A Bonferroni corrected *post-hoc* test revealed that HFD value was statistically significantly lower for the MCS (0.196 ± 0.006, $p = 0.001$, Cohen's $d = 1.66$) and VS (0.159 ± 0.007, $p < 0.001$, Cohen's $d = 3.73$) compared to the HC group (0.225 ± 0.006). There was also statistically significant difference between the MCS and VS groups ($p < 0.001$, Cohen's $d = 2.08$) see also Fig. 2 (Middle panel) and Fig. S.2.

3.4. *Entropy (E)*

E log-transformed showed a statistically significant difference between *DIAGNOSIS* as determined by the one-way ANOVA model ($F(2, 24) = 4.36$, $p = 0.0242$). A Bonferroni corrected *post-hoc* test revealed that E value was statistically significantly lower for the VS (0.799 ± 0.0459, $p = 0.0255$, Cohen's $d = 1.413$) compared to the HC group (0.837 ± 0.0192) but not between MCS (0.814 ± 0.0113) and HC group neither between MCS and VS group see also Fig. 2 (Bottom panel) and Fig. S.2.

3.5. *Is HFD more sensitive than the traditional methods to discriminate between MCS and VS?*

To compare whether the HFD was more sensitive than the PSD approach to discriminate between

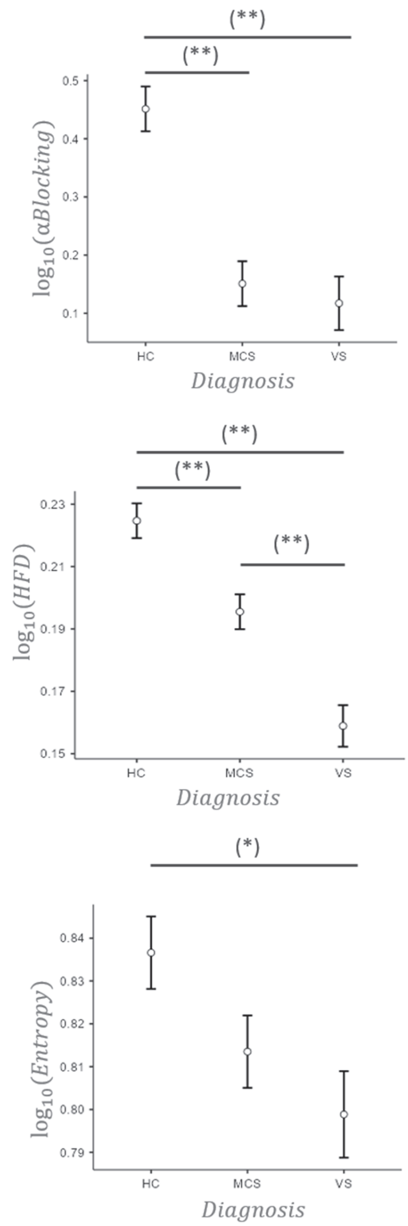

Fig. 2. αBlocking (Upper panel), HFD (Middle panel) and E (Bottom panel) log-transformed values among diagnosis (healthy controls (HC), minimally conscious state (MCS), and vegetative state (VS). The horizontal bar indicates which contrast reached the significant level at $p < 0.01$ (**) and $p < 0.05$ (*).

MCS and VS groups, we used Bayesian independent samples t-tests. There are several advantages in using Bayesian methods over traditional frequentist approaches. First, Bayesian methods allow making inferences about the null hypothesis and the alternative hypothesis. Second, it is possible to compare Bayes Factors (BF) across analyses and, based on the magnitude of the BF, derive whether one result is more robust than another.

For the HFD, a significant diagnosis (MCS versus VS) difference (Student's $t(15) = 3.78, p = 0.002$, Cohen's $dx = 1.86$, $\mathrm{BF}_{10} = 18.6$) as well as for the α band power (Student's $t(15) = 2.133$, $p = 0.05$, Cohen's $d = 1.051$, $\mathrm{BF}_{10} = 1.745$) was found. A trend, instead, was found for α/θ ratio (Student's $t(15) = 1.820$, $p = 0.089$, Cohen's $d = 0.897$,

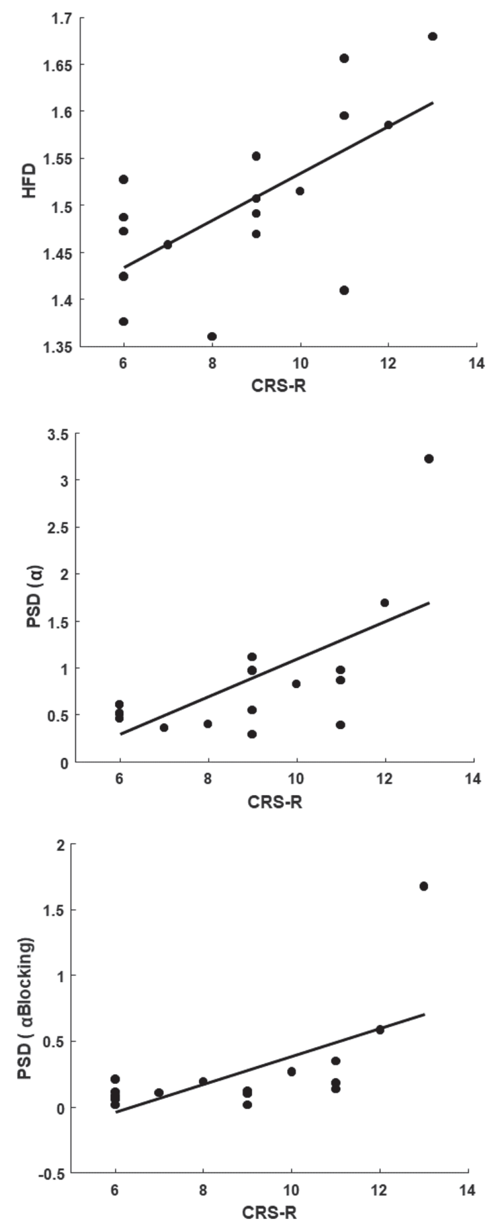

Fig. 3. Scatter plot between Coma Recovery Scale-Revised (CRS-R) and HFD (Upper panel), PSD α band (middle panel) and PSD α Blocking (Bottom panel).

$BF_{10} = 1.219$). No difference between MCS and VS was found for E(Student's $t(15) = 0.980$, $p = 0.343$, Cohen's $d = 0.483$, $BF_{10} = 0.585$).

The magnitudes of BF for group differences using HFD, PSD and E were different (18.5 for HFD compared to 1.745 for the α band, 1.219 for the α/θ ratio and 0.585 for E). This means that differences between MCS and VS were 11 times more likely to be detected by HFD compared to the α band power, 15 times with respect to the α/θ ratio and almost 32 times with respect to E.

3.6. *Correlation analysis*

Significant positive Spearman's correlation with the CRS-R clinical scale was found for α band (rho = 0.5220, $p = 0.0316$), PSD αBlocking (rho = 0.6397, $p = 0.006$) and HFD (Rho = 0.6034, $p = 0.0103$) (see Fig. 3). Interestingly, the presence of an outlier value (up right corner of the mid- and right-panel in Fig. 3) seems to condition the significance of the resulting correlation for the PSD indexes (α and αBlocking). By excluding the value associated with that subject, the correlation disappears for those indexes, but not for the HFD showing its robustness.

3.7. *Support Vector Machine (SVM)*

Moving beyond the group-level approach, the use of SVM led to an accuracy of 88.6% in discriminating the three classes. Confusion matrix is also presented in Fig. 4.

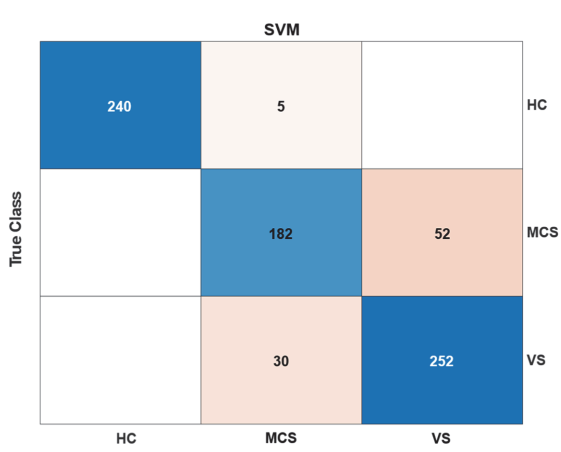

Fig. 4. Confusion matrix. The SVM showed to be very effective in distinguishing not only HC from patient populations, but also MCS from VS patients.

4. Discussion

In recent years, more and more attention has been devoted to the search for electrophysiological markers capable of supporting clinical scales to improve the diagnosis in DOC patients.[60,61] The growing interest in the use of electrophysiological techniques such as EEG compared to other neuroimaging techniques such as fMRI and PET is mainly due to the low cost of the EEG, widespread availability in the clinical setting, non-invasiveness, and portability, which makes this instrument unique for use at the patient's bedside.[60,61] Furthermore, the increasing number of electrophysiological markers developed in the last decade supports this idea.[60,61] In particular, the vast majority of these indices have been developed using linear methods based on the Fourier transform, limiting nonlinear methods to very few studies, although findings favor the latter.[62,63] In this work, we compared one of the most promising nonlinear EEG markers, the Higuchi Fractal Dimension (HFD) with the linear ones based on FFT.[13,16,17] The HFD, to our knowledge, has not yet been explored in DOC patients recorded by the EEG.

Linear methods such as PSD bands δ, θ, α, β, γ plus the α/θ ratio and α*Blocking* have certainly had the merit of tracing the route in seeking electrophysiological markers capable of discriminating between the various states of consciousness as documented by the literature.[60] Even though these methods have demonstrated an apparent capacity in discriminating between HC and DOC patients nevertheless, the same techniques have often shown mixed results in distinguishing between MCS and VS.[18,19,64–66] For example, in Liberati and colleagues,[10] it has been shown that among the 47 measurements reviewed, including frequency features at rest, 24 failed to discriminate between MCS and VS significantly.

On the one hand, some studies on rest EEG condition showed that the δ, θ, and α bands were higher in VS than in MCS.[64] On the other hand, different studies showed that the power in the α band was higher in MCS patients than in VS patients.[18,66,67] Furthermore, other works have not shown any significant differences for any of the frequency bands in the EEG resting data investigated between MCS and VS.[65] The partial agreement of the results may be attributable to the different methods used to estimate the FFT and the high sensitivity of these methods to the artefacts, strongly present in the EEG recordings, and consequently to the different approaches used to clean the EEG data from those artefacts. Last but not least, the various statistical methods used and the presence or absence of the correction for multiple comparisons in some of these studies are often ignored.

In our case, using ICA-based approaches for artefact cleaning and a very conservative approach in correction for multiple comparisons such as the Bonferroni correction, we found that the α and β bands were significantly higher for the HC to the VS group. However, they were not significantly different between HC and MCS and between MCS and VS (see Fig. 1). Furthermore, the α/θ ratio was significantly higher for the HC groups than the MCS and VS groups, but not significantly different between the MCS and VS groups (see Fig. 1). Similar results were found for the α*Blocking* index, higher for the HC group than the MCS and VS groups (see Fig. 2), but not between the MCS versus VS (see Fig. 2). Therefore, our results are consistent with those studies in which classical FFT-based linear methods can discriminate between healthy subjects and DOC patients but not between MCS and VS patients.[10,65]

In addition to linear methods, nonlinear methods such as Fractal Dimension have also been introduced as markers in both healthy[13,17,32] and pathological conditions.[14,16,42,43] The rationale, which makes non-linear methods suitable for EEG application, is linked to the fact that, in many cases, cerebral signals are considered merely oscillators with a sinusoidal basis. At the same time, it has been shown that, in some cases, a different behavior with short periods of activity that is repeated intermittently (*non-rhythmic*) is present.[27,28] For example, the rolandic μ-*rhythm* can be seen as non-sinusoidal forms characterized by one extreme (e.g. its peak) higher and more acute than the other.[23] Furthermore, the same non-sinusoidal behavior can be observed in cortical beta motor activity,[68] which instead has a more sawtooth shape that rises rapidly before falling more slowly, or vice versa.[23,68] Finally, the idea that brain rhythms are not only sinusoidal is reinforced by recent neuromodulation studies[69–72] in which it was found that modifying the stimulating waveform at 10 Hz with non-sinusoidal waveforms, such as sawtooth waves, resulted in greater efficacy of neuromodulation.[71,72] This suggests not only that non-sinusoidal stimulation are critical to entrain the network but also that non-sinusoidal oscillators synchronize faster with each other than sinusoidal oscillators.[71–73] Therefore, recognizing that physiological time series contain *hidden information* that could be captured by nonlinear methods could provide crucially and so far ignored information in DOC patients.

In DOC patients, among the nonlinear methods, entropy is predominantly used compared to the HFD, which has been reported in three very recent studies, two using fMRI data[62,63] and one using EEG-TMS.[74] fMRI studies[62,63] showed that HFD could discriminate between HC and DOC patients and MCS and VS patients. While the EEG-TMS study[74] revealed the ability of HFD to discriminate between unconscious states, including sleep and sedation induced by different anaesthetic agents. Our results, using EEG at rest, are consistent with previous literature, showing a statistically significant difference, even after Bonferroni correction, between the three groups (HC, VS, and MCS), showing higher HFD for the HC group, intermediate for the MCS group, and lower for the VS group (see Fig. 3). This aspect was further confirmed through the application of the SVM approach which enabled distinguishing the patients groups with high accuracy (Fig. 4).

Furthermore, correlation analysis between HFD and CRS-R test showed a positive correlation with higher CRS-R scores related to higher brain complexity (see Fig. 3). Even if this trend was obtained also with the linear method (PSD(α) and αBlocking indexes), the trend was driven by an outlier that if excluded, made the correlation disappear.

5. Conclusion

In conclusion, our results suggest that EEG data recorded at rest in DOC patients might be well characterized by fractal dimension analysis to more traditional FFT-based analysis, especially to discriminate between MCS and VS patients. In this respect, we have found that the HFD is 11 times more sensitive with respect to FFT-based analysis, suggesting that more complex brain dynamics at rest, as expressed by the HFD metric, might be associated with higher levels of consciousness that are not properly captured by FFT-based procedure. Moreover, machine learning showed high accuracy (88.6%) in discriminating among VS, MCS and HC using the HFD feature. Thus, with this study, we provided further evidence supporting the use of HFD for clinical application, especially to tackle misdiagnosis for those pathologies that do not show a clear symptomatic profile.

However, we are aware that the main weakness of the results obtained is the low number of subjects tested. Thus the hypothesis tested in this work needs to be confirmed in a larger cohort of patients.

Acknowledgments

CP was supported by the "Department of excellence 2018–2022" initiative of the Italian Ministry of education (MIUR) awarded to the Department of Neuroscience — University of Padua.

MM was funded by Research Foundation Flanders (FWO) (postdoctoral fellowship 1211820N).

References

1. T. Bayne, J. Hohwy and A. M. Owen, Are there levels of consciousness? *Trends Cogn. Sci.* **20** (2016) 405–413, doi:10.1016/j.tics.2016.03.009.
2. D. Habbal *et al.*, Volitional electromyographic responses in disorders of consciousness, *Brain Inj.* **28**(9) (2014) 1171–1179, doi:10.3109/02699052.2014.920519.
3. B. C. Eapen, J. Georgekutty, B. Subbarao, S. Bavishi and D. X. Cifu, *Phys. Med. Rehabil. Clin. N Am.* **28**(2) (2017) 245–258, doi:10.1016/j.pmr.2016.12.003.
4. S. Laureys *et al.*, Auditory processing in the vegetative state, *Brain* **123**(8) (2000) 1589–1601, doi:10.1093/brain/123.8.1589.
5. S. R. Deputy, The minimally conscious state in children, *Clin. Pediatr.* (*Phila.*). **41**(9) (2002) 735–736, doi:10.1177/000992280204100917.
6. M. Boly *et al.*, Functional connectivity in the default network during resting state is preserved in a vegetative but not in a brain dead patient, *Hum. Brain Mapp.* **30**(8) (2009) 2393–2400, doi:10.1002/hbm.20672.
7. M. A. Bruno *et al.*, Disorders of consciousness: Moving from passive to resting state and active paradigms, *Cogn. Neurosci.* **1**(3) (2010) 193–203, doi:10.1080/17588928.2010.485677.
8. T. Bekinschtein *et al.*, Emotion processing in the minimally conscious state [1], *J. Neurol. Neurosurg. Psychiatry.* **75**(5) (2004) 788, doi:10.1136/jnnp.2003.034876.
9. C. Schnakers *et al.*, Diagnostic accuracy of the vegetative and minimally conscious state: Clinical consensus versus standardized neurobehavioral assessment, *BMC Neurol.* **9** (2009) 35, doi:10.1186/1471-2377-9-35.
10. G. Liberati, T. Hünefeldt and M. O. Belardinelli, Questioning the dichotomy between vegetative state and minimally conscious state: A review of the statistical evidence, *Front. Hum. Neurosci.* **8**(November) (2014) 1–14, doi:10.3389/fnhum.2014.00865.
11. J. C. O'Donnell, K. D. Browne, T. J. Kilbaugh, H. I. Chen, J. Whyte and D. K. Cullen, Challenges and demand for modeling disorders of consciousness following traumatic brain injury, *Neurosci Biobehav Rev.* **98** (2019) 336–346, doi:10.1016/j.neubiorev.2018.12.015.
12. J. Samogin *et al.*, Frequency-dependent functional connectivity in resting state networks, *Hum. Brain Mapp.* **41**(18) (2020) 5187–5198, doi:10.1002/hbm.25184.
13. M. Marino *et al.*, Neuronal dynamics enable the functional differentiation of resting state networks in the human brain, *Hum. Brain Mapp.* **40**(5) (2019) 1445–1457, doi:10.1002/hbm.24458.
14. F. M. Smits, C. Porcaro, C. Cottone, A. Cancelli, P. M. Rossini and F. Tecchio, Electroencephalographic fractal dimension in healthy ageing and Alzheimer's disease, *PLoS One.* **11**(2) (2016) e0149587, doi:10.1371/journal.pone.0149587.
15. S. Carletto, C. Porcaro, C. Settanta *et al.*, Neurobiological features and response to eye movement desensitization and reprocessing treatment of posttraumatic stress disorder in patients with breast cancer, *Eur. J. Psychotraumatol.* **10**(1) (2019) 1600832, doi:10.1080/20008198.2019.1600832.
16. C. Porcaro, C. Cottone, A. Cancelli, P. M. Rossini, G. Zito and F. Tecchio, Cortical neurodynamics changes mediate the efficacy of a personalized neuromodulation against multiple sclerosis fatigue, *Sci. Rep.* **9**(1) (2019) 1–10, doi:10.1038/s41598-019-54595-z.
17. C. Cottone, C. Porcaro, A. Cancelli, E. Olejarczyk, C. Salustri and F. Tecchio, Neuronal electrical ongoing activity as a signature of cortical areas, *Brain Struct. Funct.* **222**(5) (2017) 2115–2126, doi:10.1007/s00429-016-1328-4.
18. R. Lehembre *et al.*, Resting-state EEG study of comatose patients: A connectivity and frequency analysis to find differences between vegetative and minimally conscious states, *Funct. Neurol.* **27**(1) (2012) 41–47. https:// pubmed.ncbi.nlm.nih.gov/22687166/ [accessed on 13th October 2021].
19. R. Lehembre *et al.*, Electrophysiological investigations of brain function in coma, vegetative and minimally conscious patients, *Arch. Ital. Biol.* **150**(2–3) (2012) 122–139, doi:10.4449/aib.v150i2.1374.
20. W. Klimesch, B. Schack, M. Schabus, M. Doppelmayr, W. Gruber and P. Sauseng, Phase-locked alpha and theta oscillations generate the P1-N1

complex and are related to memory performance, *Cogn. Brain Res.* **19**(3) (2004) 302–316, doi:10.1016/j.cogbrainres.2003.11.016.
21. M. T. Schmidt *et al.*, Index of alpha/theta ratio of the electroencephalogram: A new marker for Alzheimer's disease, *Front. Aging Neurosci.* **5** (2013) 60, doi:10.3389/fnagi.2013.00060.
22. N. D. Schiff, Recovery of consciousness after brain injury: A mesocircuit hypothesis, *Trends Neurosci.* **33**(1) (2010) 1–9, doi:10.1016/j.tins.2009.11.002.
23. S. R. Cole and B. Voytek, Brain oscillations and the importance of waveform shape, *Trends Cogn. Sci.* **21**(2) (2017) 137–149, doi:10.1016/j.tics.2016.12.008.
24. S. R. Jones, When brain rhythms aren't 'rhythmic': Implication for their mechanisms and meaning, *Curr. Opin. Neurobiol.* **40** (2016) 72–80, doi:10.1016/j.conb.2016.06.010.
25. C. J. Stam, Nonlinear dynamical analysis of EEG and MEG: Review of an emerging field, *Clin. Neurophysiol.* **116**(10) (2005) 2266–2301. doi:10.1016/j.clinph.2005.06.011.
26. W. Klonowski, Everything you wanted to ask about EEG but were afraid to get the right answer, *Nonlinear Biomed. Phys.* **3** (2005) 1–5, doi:10.1186/1753-4631-3-2.
27. J. Feingold, D. J. Gibson, B. Depasquale and A. M. Graybiel, Bursts of beta oscillation differentiate postperformance activity in the striatum and motor cortex of monkeys performing movement tasks, *Proc. Natl. Acad. Sci. U. S. A.* **112**(44) (2015) 13687–13692, doi:10.1073/pnas.1517629112.
28. M. Lundqvist, J. Rose, P. Herman, S. L. L. Brincat, T. J. J. Buschman and E. K. K. Miller, Gamma and beta bursts underlie working memory, *Neuron.* **90**(1) (2016) 152–164, doi:10.1016/j.neuron.2016.02.028.
29. A. L. Goldberger, Heartbeats, hormones, and health: Is variability the spice of life? *Am. J. Respir. Crit. Care Med.* **163**(6) (2001) 1289–1290, doi:10.1164/ajrccm.163.6.ed1801a.
30. A. L. Goldberger, L. A. N. Amaral, J. M. Hausdorff, P. C. Ivanov, C. K. Peng and H. E. Stanley, Fractal dynamics in physiology: Alterations with disease and aging, *Proc. Natl. Acad. Sci. USA* **99**(SUPPL. 1) (2002) 2466–2472, doi:10.1073/pnas.012579499.
31. W. Klonowski, From conformons to human brains: An informal overview of nonlinear dynamics and its applications in biomedicine, *Nonlinear Biomed. Phys.* **1** (2007) 1–19, doi:10.1186/1753-4631-1-5.
32. C. Porcaro, S. D. Mayhew, M. Marino, D. Mantini and A. P. Bagshaw, Characterisation of haemodynamic activity in resting state networks by fractal analysis, *Int. J. Neural Syst.* **30**(12) (2020) S0129065720500616, doi:10.1142/S0129065720500616.
33. G. Rodríguez-Bermúdez and P. J. García-Laencina, Analysis of EEG signals using nonlinear dynamics and chaos: A review, *Appl. Math. Inf. Sci.* **9**(5) (2015) 2309–2321, doi:10.12785/amis/090512.
34. D. Zhang and M. E. Raichle, Disease and the brain's dark energy, *Nat. Rev. Neurol.* **6**(1) (2010) 15–28, doi:10.1038/nrneurol.2009.198.
35. A. Di Ieva, F. Grizzi, H. Jelinek, A. J. Pellionisz and G. A. Losa, Fractals in the neurosciences, part I: General principles and basic neurosciences, *Neuroscientist.* **20**(4) (2014) 403–417, doi:10.1177/1073858413513927.
36. A. Di Ieva, F. J. Esteban, F. Grizzi, W. Klonowski, M. Martín-Landrove, Fractals in the neurosciences, Part II: Clinical applications and future perspectives, *Neuroscientist.* **21**(1) (2015) 30–43, doi:10.1177/1073858413513928.
37. M. Ahmadlou and H. Adeli, Visibility graph similarity: A new measure of generalized synchronization in coupled dynamic systems, *Phys. D Nonlinear Phenomena* **241**(4) (2012) 326–332.
38. M. Ahmadlou, H. Adeli and A. Adeli, Fractality and a wavelet-chaos-neural network methodology for EEG-based diagnosis of autistic spectrum disorder, *J. Clin. Neurophysiol.*, Published online 2010, doi:10.1097/WNP.0b013e3181f40dc8.
39. M. Ahmadlou, H. Adeli and A. Adeli, Fractality analysis of frontal brain in major depressive disorder, *Int. J. Psychophysiol.* **85**(2) (2012) 206–211, doi:10.1016/j.ijpsycho.2012.05.001.
40. M. Ahmadlou, H. Adeli and A. Adeli, Improved visibility graph fractality with application for the diagnosis of autism spectrum disorder, *Phys. A Stat. Mech. its Appl.* **391**(20) (2012) 4720–4726, doi:10.1016/j.physa.2012.04.025.
41. H. Adeli, S. Ghosh-Dastidar and N. Dadmehr, A spatio-temporal wavelet-chaos methodology for EEG-based diagnosis of Alzheimer's disease, *Neurosci. Lett.* **444**(2) (2008) 190–194, doi:10.1016/j.neulet.2008.08.008.
42. C. Porcaro, A. Di Renzo, E. Tinelli, G. Di Lorenzo, V. Parisi, F. Caramia, M. Fiorelli, V. Di Piero, F. Pierelli and G. Coppola, Haemodynamic activity characterization of resting state networks by fractal analysis and thalamocortical morphofunctional integrity in chronic migraine, *J. Headache Pain.* **21**(1) (2020) 112, doi:10.1186/s10194-020-01181-8. PMID: 32928129; PMCID: PMC 7490862.
43. C. Porcaro, A. Di Renzo, E. Tinelli, G. Di Lorenzo, S. Seri, C. Di Lorenzo, V. Parisi, F. Caramia, M. Fiorelli, V. Di Piero, F. Pierelli, G. Coppola, Hypothalamic structural integrity and temporal complexity of cortical information processing at rest in migraine without aura patients between attacks, *Sci. Rep.* **11**(1) (2021) 18701, doi:10.1038/s41598-021-98213-3.
44. C. Cattani and G. Pierro, On the fractal geometry of DNA by the binary image analysis, *Bull. Math.*

Biol. **75**(9) (2013) 1544–1570, doi:10.1007/s11538-013-9859-9.

45. C. Y. Lee, The fractal dimension as a measure for characterizing genetic variation of the human genome, *Comput. Biol. Chem.* **87** (2020) 107278, doi:10.1016/j.compbiolchem.2020.107278.
46. A. Borri, A. Cerasa, P. Tonin, L. Citrigno and C. Porcaro, Characterizing fractal genetic variation in the human genome from the Hapmap project, *Int J Neural Syst.* **12** (2022) 2250028, doi:10.1142/S0129065722500289.
47. T. Higuchi, Approach to an irregular time series on the basis of the fractal theory, *Phys. D Nonlinear Phenom.* **31**(2) (1988) 277–283, doi:10.1016/0167-2789(88)90081-4.
48. O. A. Rosso, Entropy changes in brain function, *Int. J. Psychophysiol.* **64**(1) (2007) 75–80, doi:10.1016/j.ijpsycho.2006.07.010.
49. J. T. Giacino, K. Kalmar and J. Whyte, The JFK coma recovery scale-revised: Measurement characteristics and diagnostic utility, *Arch. Phys. Med. Rehabil.* **85**(12) (2004) 2020–2029, doi:10.1016/j.apmr.2004.02.033.
50. C. Porcaro, M. T. Medaglia and A. Krott, Removing speech artifacts from electroencephalographic recordings during overt picture naming, *Neuroimage.* **105** (2015) 171–180. doi:10.1016/j.neuroimage.2014.10.049.
51. G. Barbati, C. Porcaro, F. Zappasodi, P. M. Rossini and F. Tecchio, Optimization of an independent component analysis approach for artifact identification and removal in magnetoencephalographic signals, *Clin Neurophysiol.* **115**(5) (2004) 1220–1232, doi:10.1016/j.clinph.2003.12.015.
52. D. L. Schomer and F. H. L. da Silva, Niedermeyer's electroencephalography: Basic principles, in *Clinical Applications, and Related Fields*, 6th edn. (2012), Ernst Niedermeyer, F. H. Lopes da Silva Lippincott Williams & Wilkins, doi:10.1111/j.1468-1331.2011.03406.x.
53. C. Ramon, M. D. Holmes, Spatiotemporal phase clusters and phase synchronization patterns derived from high density EEG and ECoG recordings, *Curr. Opin. Neurobiol.* **31** (2015) 127–132. doi:10.1016/j.conb.2014.10.001.
54. C. E. Shannon, A mathematical theory of communication, *Bell Syst. Tech. J.* **27** (1948) pp. 379423, 623656, doi:10.1002/j.1538-7305.1948.tb01338.x.
55. IFSECN, A glossary of terms most commonly used by clinical electroencephalographers, *Electroencephalogr Clin Neurophysiol.* **37**(5) (1974) 538–548, doi:10.1016/0013-4694(74)90099-6.
56. M. X. Cohen, Comparison of different spatial transformations applied to EEG data: A case study of error processing, *Int. J. Psychophysiol.*, Published online 2015, doi:10.1016/j.ijpsycho.2014.09.013.
57. D. Ostwald, C. Porcaro and A. P. Bagshaw, An information theoretic approach to EEG-fMRI integration of visually evoked responses, *Neuroimage.* **49**(1) (2010) 498–516, doi:10.1016/j. neuroimage.2009.07.038.
58. D. Ostwald, C. Porcaro and A. P. Bagshaw, Voxel-wise information theoretic EEG-fMRI feature integration, *Neuroimage.* **55**(3) (2011) 1270–1286, doi:10.1016/j.neuroimage.2010.12.029.
59. T. E. Nichols and A. P. Holmes, Nonparametric permutation tests for functional neuroimaging: A primer with examples, *Hum. Brain Mapp.* **15**(1) (2002) 1–25, doi:10.1002/hbm.1058.
60. A. Comanducci *et al.*, Clinical and advanced neurophysiology in the prognostic and diagnostic evaluation of disorders of consciousness: Review of an IFCN-endorsed expert group, *Clin. Neurophysiol.* **131**(11) (2020) 2736–2765, doi:10.1016/j.clinph.2020.07.015.
61. C. Porcaro, I. E. Nemirovsky, F. Riganello, Z. Mansour, A. Cerasa, P. Tonin, B. Stojanoski and A. Soddu, Diagnostic Developments in Differentiating Unresponsive Wakefulness Syndrome and the Minimally Conscious State, *Front Neurol.* **12** (2022) 778951, doi:10.3389/fneur.2021.778951.
62. T. F. Varley, M. Craig, R. Adapa, P. Finoia, G. Williams, J. Allanson, J. Pickard, D. K. Menon and E. A. Stamatakis, Fractal dimension of cortical functional connectivity networks & severity of disorders of consciousness, *PLoS One.* **15**(2) (2020) e0223812, doi:10.1371/journal.pone.0223812.
63. A. I. Luppi, M. M. Craig, P. Coppola, A. R. D. Peattie, P. Finoia, G. B. Williams, J. Allanson, J. D. Pickard, D. K. Menon, E. A. Stamatakis, Preserved fractal character of structural brain networks is associated with covert consciousness after severe brain injury, *Neuroimage Clin.* **30** (2021) 102682, doi:10.1016/j.nicl.2021.102682.
64. A. A. Fingelkurts, A. A. Fingelkurts, S. Bagnato, C. Boccagni, G. Galardi, The value of spontaneous EEG oscillations in distinguishing patients in vegetative and minimally conscious states. *Suppl. Clin. Neurophysiol.* **62** (2013) 81–99, doi:10.1016/b978-0-7020-5307-8.00005-3.
65. J. Lechinger *et al.*, Mirroring of a simple motor behavior in disorders of consciousness, *Clin. Neurophysiol.* **124**(1) (2013) 27–34, doi:10.1016/j.clinph.2012.05.016.
66. F. Riganello, M. Vatrano, S. Carozzo, M. Russo, L. F. Lucca, M. Ursino, V. Ruggiero, A. Cerasa and C. Porcaro, The Timecourse of Electrophysiological Brain-Heart Interaction in DoC Patients, *Brain Sci.* **11**(6) (2021) 750, doi:10.3390/brainsci11060750.
67. C. Babiloni *et al.*, Cortical sources of resting-state alpha rhythms are abnormal in persistent vegetative state patients, *Clin. Neurophysiol.* **120**(4) (2009) 719–729, doi:10.1016/j.clinph.2009.02.157.

68. S. R. Cole, R. van der Meij, E. J. Peterson, C. de Hemptinne, P. A. Starr and B. Voytek, Nonsinusoidal Beta Oscillations Reflect Cortical Pathophysiology in Parkinson's Disease, *J. Neurosci.* **37**(18) (2017) 4830–4840, doi:10.1523/JNEUROSCI.2208-16.2017.
69. C. Cottone, A. Cancelli, P. Pasqualetti, C. Porcaro, C. Salustri and F. Tecchio, A new, high-efficacy, noninvasive transcranial electric stimulation tuned to local neurodynamics, *J. Neurosci.* **38**(3) (2018) 586–594, doi:10.1523/JNEUROSCI.2521-16.2017.
70. F. Fröhlich, Experiments and models of cortical oscillations as a target for noninvasive brain stimulation, *Prog. Brain Res.* **222** (2015) 41–73, doi:10.1016/bs.pbr.2015.07.025.
71. F. Fröhlich and D. A. McCormick, Endogenous electric fields may guide neocortical network activity, *Neuron.* **67**(1) (2010) 129–143, doi:10.1016/j.neuron.2010.06.005.
72. J. Dowsett and C. S. Herrmann, Transcranial alternating current stimulation with sawtooth waves: Simultaneous stimulation and EEG recording, *Front. Hum. Neurosci.* **10** (2016) 1–10, doi:10.3389/fnhum.2016.00135.
73. D. Somers and N. Kopell, Rapid synchronization through fast threshold modulation, *Biol Cybern.* **68**(5) (1993) 393–407, doi:10.1007/BF00198772. PMID: 8476980.
74. J. Ruiz de Miras *et al.*, Fractal dimension analysis of states of consciousness and unconsciousness using transcranial magnetic stimulation, *Comput. Methods Programs Biomed.* **175** (2019) 129–137, doi:10.1016/j.cmpb.2019.04.017.

Appendix A. Supplementary Figures

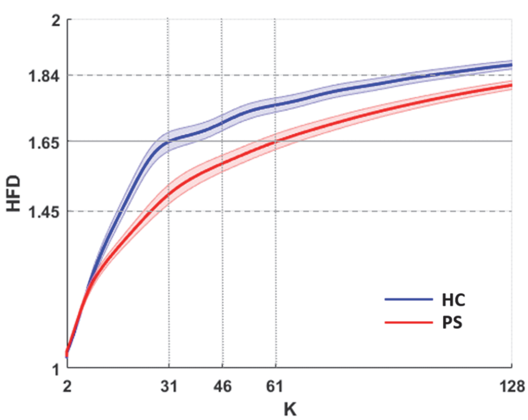

Fig. S.1. (Color online) Higuchi's Fractal Dimension (HFD) estimate for all possible k values was performed on Healthy Controls (HC – blue line) and Pathological Subjects (PS – red line (i.e. MCS plus VS)). The blue and red lines are the averaged FD values across HCs and PSs, respectively, and the shadow areas indicate the standard error (SE). The horizontal grey line indicates the median value of the averaged HCs and PSs, whereas the dashed horizontal grey lines indicate median value $\pm$ standard deviation. Dotted lines at $k = 31$ and $k = 61$ represent when the median value intercepts the HC and PS HFD values. The dotted line at $k = 46$ represents the k value used in our study.

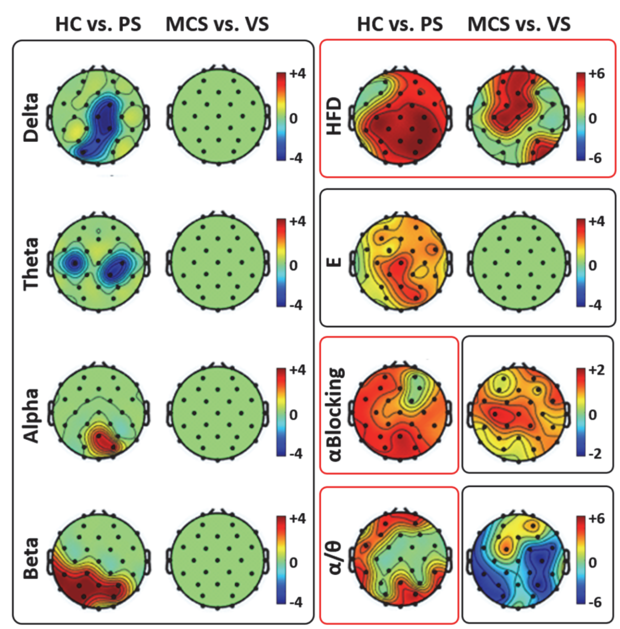

Fig. S.2. (Color online) Topographical representation of the non-parametric t-test between HC versus PS and MCS versus VS contrasts for all the EEG features extracted. The results shown in the red square represents the tests that survived the FDR correction. All the others are not FDR corrected. Colorbar represents the T-Value of the non-parametric t-test.